

Exercise 6 NULL Functions

Question 1

SELECT

employ-id,

name,

COALESCE (salary, 0) as salary-with-default

FROM employees;

COALESCE (salary, 0) returns salary if not NULL, otherwise 0

employ-id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2 Show employee names with their department, if Null show "Not Assigned"

SELECT

employee_id,

name,

COALESCE (department, "Not Assigned") AS department_name,

FROM Employees;

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Q3

Q3

Find orders with NULL customer-id using ISNULL

SELECT

order-id,

customer-id,

FROM orders WHERE ISNULL (customer-id, 0) = 0;

Explanation : ISNULL (customer-id, 0) replaces NULL with 0, so we filter rows where customer-id was NULL.

Sy

Show all orders, if delivery-date is Null show 'Pending'

SELECT

order-id,

customer-id,

COALESCE (CAST (delivery-date AS VARCHAR),

order-id	customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	Null	2024-12-03

Q5

Show all orders, if delivery-date is Null show 'Pending'

103 | Null | 2024-12-05

Q5

Show all students & their grades! replace Null with 0

SELECT

student_id,

name,

COALESCE (grade, 0) AS final_grade,
FROM students;

student_id

name

final_grade

1

Ethan

85

2

Maya

0

3

Olivia

90

3	Olivia	90
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SELECT COUNT(*) AS not-graded-count,
FROM Student,
WHERE grade IS NULL;

Row Result	not-graded-count
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Q7
Products: Show product-id, name & final-price after discount (assume discount 0, if null)

* Discount is amount subtracted.

SELECT
product-id,
name,
price - COALESCE(discount, 0) AS final-price
FROM Products;

Product-id	name	
S01	Keyboard	25
S02	Mouse	10
S03	Monitor	100

Q8 - Can't know how many customers have no email.
(Can't use COUNT(email_count), as it will count nulls as 0)

SO 2

Mouse

10

SO 3

Monitor

100

Qs - Count how ~~no~~ many customers have no email.

```
SELECT COUNT(*) AS missing-email-count,  
FROM Customers,  
WHERE email IS NULL;
```

Results

2

Question 9 Show all customers with email, if NULL display 'No email'

SELECT

customer_id,

name,

~~COE~~ COALESCE (email, 'No email') AS email_display
FROM customers;

Result

customer_id	name	email_display
1	Linda	No email
2	Joseph	Joseph@amazon
3	N/A	No email

N/A

2	Joseph	joseph@mail.com
3	N/A	No mail

Q10

Payment: Show payment details with method replaced by "unknown" if null

SELECT

payment-id,
COALESCE (method, 'unknown')

payment-id	method-display	Status
301	Credit	Null
302	Paypal	Success
303	Unknown	Failed

Q
Inventory: Show items & quantity (if Null)

SELECT

Item-id,

Item-name,
COALESCE (quantity) AS quantity-checked,
FROM Inventory;
EDA

Item-id	Item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Erase	0

3

Eraser

0

Q12

Employee Extra: show 1 first available value among

```
SELECT emp_id,  
COALESCE (bonus, commission) AS first-available-reward  
FROM employees Extra;
```

emp_id	First-available-reward
1	300
2	100
3	NULL

Q3
Count classes that don't have a room assigned

SELECT (COUNT(*) AS ~~no~~ no-of-room-count,
FROM Classes
WHERE room IS NULL;

no-room-count	2
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Q4

Attendance:

Instruction: Show records with status! replace null with "Not Marked"

Question 14

Goal Attendance:

Instructions: Show records with status; replace NULL with "Not Marked"

SELECT

student_id,

date,

COALESCE(status, "Not Marked") AS attendance_status

FROM Attendance;

Student_id	Date	Attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Question 15

SELECT

account-id,

COALESCE(account-type, 'unknown') AS type-display,

COALESCE(balance) AS balance-checked

FROM Bank-Accounts;

account-id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Die

Projects: Show all

SELECT

project-id,

title,

COALESCE(CAST(start_date AS VARCHAR), 'TBD') AS start_display
FROM projects;

project-id	title	start_display
1	Website	2025-01-01
2	Mobile	TBD
3	Data Migration	TBD